

human language users. This distinction forms the critical foundation for the fallacies examined throughout this paper.

Interpretive Fallacies in LLM-Based Research

To understand why LLMs function as simulators rather than replacements, we must first examine the conceptual errors that underlie the replacement perspective. Six fallacies emerge in discussions of LLMs as human substitutes, each revealing a different way researchers might misinterpret what LLMs are and what their outputs signify (**Table 1**).

Table 1

Six Fallacies in Conceptualizing LLMs as Human Replacements

Fallacy	Core Misconception	Why It's Problematic	Research Implications
Token prediction as intelligence	Mistaking statistical pattern matching for human understanding	LLMs process language without embodied experience, consciousness, or genuine understanding	Cannot assume cognitive equivalence or generalize from LLM performance to human cognition
The average human	Assuming LLMs represent typical or average human responses	Training biases (WEIRD, temporal) and optimization goals distort LLM output representation	Cannot reliably generalize findings to diverse human populations or contexts
Alignment as explanation	Inferring shared cognitive mechanisms from similar outputs	Different architectures can produce similar results (multiple realizability); surface similarity is not mechanistic equivalence	Output alignment does not confirm shared underlying processes with humans
Anthropomorphism	Attributing human-like mental states to statistical systems	Current LLMs lack consciousness, intentions, or genuine emotional understanding	Invalidates psychological constructs requiring mental states; risks misinterpretation
Identity essentialization	Treating demographic labels as fixed, homogeneous categories	Reduces complex, fluid, and intersectional identities to stereotypes; ignores within-group diversity	Misrepresents population diversity and the construct of identity itself
Substitution	Using LLMs as primary tools for understanding human cognition or behavior	Static training data, lack of real-time adaptability, and potential for closed-loop effects	LLM data requires human validation; cannot directly replace human participants

Token Prediction as Human Intelligence Fallacy

LLMs accomplish tasks once reserved for humans through a layered process: They are initially pretrained to predict sequential tokens; then post-trained via fine-tuning and reinforcement learning; and finally, their reasoning behavior is shaped by guidance from well-crafted prompts and inference-time techniques (**Box 1**). This process equips LLMs with intricate linguistic knowledge (e.g., syntactic rules, semantic relations)—often termed *formal* linguistic competence (Mahowald et al., 2024). Crucially, these models do more than

memorize patterns; they demonstrate the ability to navigate complex, context-dependent tasks in both language processing and generation (Mahowald et al., 2024; Milli re & Buckner, 2024a). Through learning and contextual adaptation, LLMs can infer underlying task structures and generate appropriate responses across varied domains—a form of *instrumental* knowledge that supports problem-solving in diverse settings (Yildirim & Paul, 2024).

Yet, at their core, LLMs have no minds (Searle, 1980) but are autoregressive statistical models that manipulate language—a task different from human cognition, producing a kind of ungrounded intelligence. They manipulate language by leveraging statistical patterns learned during training; while these patterns encode aspects of *derivative* meaning present in text-corpus relationships (e.g., “Paris”—“France,” “fire”—“hot”), the models themselves are yet to access this meaning through direct, grounded experience (e.g., visual qualia, proprioception). This mirrors Church encoding in lambda calculus, where data and operations are expressed purely through abstract functions, defined by how they transform inputs to outputs, rather than by inherent, grounded meaning (Church, 1936). This is in contrast to human intelligence, which integrates general and specialized capabilities through evolutionary adaptations and developmental learning (**Box 2**)—and is “grounded in one’s embodied physical and emotional experiences” and “deeply reliant on one’s social and cultural environments” (Mitchell, 2024).

A fallacy emerges when we mistake the capacity of LLMs to predict language patterns and solve cognitive problems for genuine human-like understanding or intelligence. To unpack this fallacy, it is helpful to distinguish between two levels of understanding: functional understanding (reliably producing appropriate outputs, as when a dog successfully catches a ball) and reflective understanding (consciously grasping underlying principles, as when humans can articulate why the ball follows a particular trajectory). While LLMs may demonstrate considerable functional understanding by generating contextually appropriate responses, they currently lack the reflective understanding that emerges from embodied experience and consciousness. Consequently, mistaking their output for genuine cognition would misrepresent what is being measured; the model’s linguistic performance, based on statistical token prediction, risks being conflated with human cognitive processes that integrate both functional and reflective understanding. The “intelligence” of LLMs is thus operationally different from the human cognition it might appear to simulate.

This fallacy manifests in two major ways. First, while LLMs process vast amounts of text reflecting real-world information and human experience, their tokens operate without direct, first-hand sensorimotor contact with the world. Consequently, they lack embodied referents and lived experience derived from such interaction (Leivada et al., 2023). This critical absence of grounding is demonstrated by how text-based LLMs, while capable of representing non-sensorimotor features of human concepts (e.g., emotional valence), systematically fail to capture sensorimotor features, particularly those related to motor actions. Indeed, the alignment between LLM and human conceptual representations diminishes markedly from non-sensorimotor to sensorimotor domains (Xu et al., 2025).

Second, this absence of first-person, sensorimotor grounding limits the models’ connection to the physical and social realities they aim to simulate. This hinders the development of fully *functional* linguistic competence—the use of language to achieve goals in the world (Mahowald et al., 2024). It also restricts their capacity to acquire the kind of *worldly* knowledge needed for robust world models—internal representations that are structure-preserving and behaviorally efficacious in real-world interactions (Webb et al., 2023; Yildirim & Paul, 2024). Such models, including cognitive maps, body schemas, and spatial schemas, support embodied reasoning and action, which remain beyond the scope of current models (Wicke & Wachowiak, 2024).

A critical consequence of this fallacy is that when researchers attribute changes in model outputs to experimental manipulations without recognizing the underlying statistical nature, they inevitably risk misinterpretation. Consider a simple prompt: “*Mao Zedong was...*” Unlike interacting with another mind, when we engage with a chatbot, we are not seeking its opinion—despite the compelling illusion thereof—but rather making a *computational* request: Given the statistical distributions in the language model, what sequence is most likely to follow these words? Models trained for neutrality will likely provide correspondingly factual responses. Indeed, the character of such responses is shaped by the model’s training, geopolitical alignment, and prompt language. For example, DeepSeek-R1—pretrained mostly on English and Chinese text, and subsequently aligned with government regulatory directives (**Box 1**)—showed substantially higher proportions of Chinese-state propaganda and anti-U.S. bias compared with ChatGPT o3-mini-high, which was pretrained primarily on English text. This bias was most pronounced with Simplified Chinese queries, diminished with Traditional Chinese inputs, and was nearly absent when queried in English (Huang et al., 2025).

Fundamentally then, the model, unlike a human, has no communicative intent, no opinion of, attitude toward, or belief about Mao, and no intrinsic capacity to tell the truth—it just models a distribution of token sequences based on the training texts (Shanahan, 2024). While they manifest proficient language *use* that goes beyond simply retrieving pre-recorded text strings, they are not true language *users* in the philosophical sense. They do not possess intrinsic meaning, communicative intentions, or other internal states essential to human language users (Block, 1981).

Indeed, model outputs can be highly sensitive to seemingly trivial prompt variations—variations that human language users would tolerate (Ivanova, 2025). Small changes in prompt language, wording, order, or context can produce drastically different outputs not because of the manipulated variable of interest, but due to the model’s inherent sensitivity patterns—a phenomenon that contrasts with how humans process language.

Additionally, findings from LLM studies cannot be reliably generalized to human cognition. The gap between token prediction mechanisms and human cognitive processes fundamentally limits what we can infer about real human populations from model behavior, even when performance patterns may appear superficially similar. As philosopher John Searle illustrated in his “Chinese Room” thought experiment, a system might process symbols according to rules without understanding their meaning—just as a person following instructions to manipulate Chinese characters could produce appropriate responses without comprehending Chinese. When models generate coherent language without the intentionality, consciousness, and direct grounding in real-world experience that characterize human intelligence (Searle, 1980), they are capturing linguistic *form* (the observable structure of language) rather than true *meaning* (the communicative intent behind language) (Bender & Koller, 2020). Therefore, to confuse cognitive algorithms for cognition, or models of the mind for the mind itself, represents a category mistake, both ontologically (misidentifying their nature) and epistemologically (misunderstanding their knowledge).

No doubt LLMs (and AI in general) will continue to advance. But, as detailed in **Box 3**, even these improvements face fundamental conceptual limitations regarding embodied experience and grounding. Knowledge, as constructivists like Jean Piaget have argued, is built from sensory experiences and perceptions, which are then layered with symbols and categories over time. Without such experiences, LLMs are like an artificial version of the proverbial Mary studying everything about color in a black-and-white room her entire life (Jackson, 1982). One can argue that when Mary leaves her room and sees color for the first time, she learns something new—what it is like to see something pink (Jackson, 1986). If so, a great deal more is at stake for language models, which experience neither color nor